Figure 22-5 illustrates the surface of a compact laser disc, such as viewed through a high power microscope. The main surface is shiny (reflective of light), with the digital information stored as a series of dark pits burned on the surface with a laser. The information is arranged in a single track that spirals from the outside to the inside, the same as a phonograph record. The rotation of the CD is changed from about 210 to 480 rpm as the information is read from the outside to the inside of the spiral, making the scanning velocity a constant 1.2 meters per second. (In comparison, phonograph records spin at a fixed rate, such as 33, 45 or 78 rpm). During playback, an optical sensor detects if the surface is reflective or nonreflective, generating the corresponding binary information.

As shown by the geometry in Fig. 22-5, the CD stores about 1 bit per $(\mu m)^2$, corresponding to 1 million bits per $(mm)^2$, and 15 billion bits per disk. This is about the same feature size used in integrated circuit manufacturing, and for a good reason. One of the properties of light is that it cannot be focused to smaller than about one-half wavelength, or 0.3 µm. Since both integrated circuits and laser disks are created by optical means, the fuzziness of light below 0.3 µm limits how small of features can be used.

Figure 22-6 shows a block diagram of a typical compact disc playback system. The raw data rate is 4.3 million bits per second, corresponding to 1 bit each 0.28 µm of track length. However, this is in conflict with the specified geometry of the CD; each pit must be no shorter than 0.8 µm, and no longer than 3.5 µm. In other words, each binary *one* must be part of a group of 3 to 13 *ones*. This has the advantage of reducing the error rate due to the optical pickup, but how do you force the binary data to comply with this strange bunching?

The answer is an encoding scheme called **eight-to-fourteen modulation (EFM)**. Instead of directly storing a byte of data on the disc, the 8 bits are passed through a look-up table that pops out 14 bits. These 14 bits have the desired bunching characteristics, and are stored on the laser disc. Upon playback, the binary values read from the disc are passed through the inverse of the EFM look-up table, resulting in each 14 bit group being turned back into the correct 8 bits.

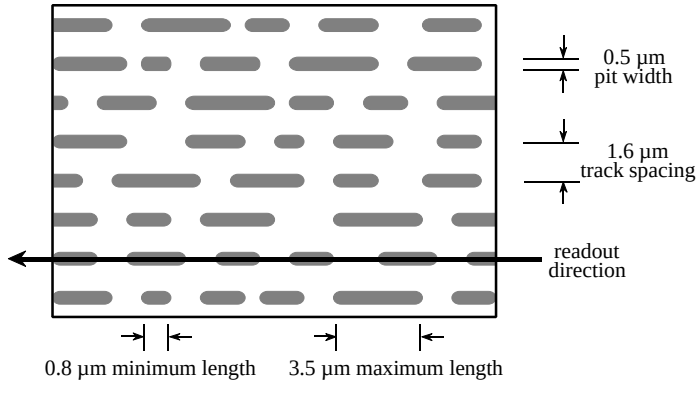

FIGURE 22-5
Compact disc surface. Micron size pits are burned into the surface of the CD to represent ones and zeros. This results in a data density of 1 bit per μm^2, or one million bits per mm^2. The pit depth is 0.16 µm.